K Bhaskar Chari

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Summary

IEEE-published final-year ML Engineer with hands-on experience across **production RAG pipelines**, **multimodal generative AI**, and **LLM fine-tuning**. Comfortable owning a system end-to-end — from data pipeline and model training through API design, observability, and test coverage. Looking for roles where **reliability and evaluation rigour** are taken as seriously as raw model performance.

Publications

Cultural and Sarcastic Context-Aware Sentiment Analysis with LLMs and Rule-Based Heuristics *IEEE CSNT, 2026*

- Designed a **hybrid dual-path architecture** routing inputs between VADER (<5ms) and **QLoRA fine-tuned LLaMA-2** via confidence threshold (τ=0.75), with cultural metadata injection and **Chain-of-Thought prompts** to resolve sarcasm and culture-specific idioms.
- Outperformed baseline BERT and VADER on the SARC dataset; each component of the architecture contributed independently to accuracy gains, with the full hybrid system achieving the highest F1 across all evaluated configurations.

Projects

DocuMind AI | FastAPI, AsyncQdrantClient, Gemini Flash, SentenceTransformers, spaCy, Docker

- Architected a **fully async RAG pipeline** offloading embedding and Gemini calls via `asyncio.to_thread()`, achieving **20% reduction** in query latency; enforced 100% source-pinned output via a **Confidence Scoring Engine** with Pydantic-validated schemas and **72 unit tests**.
- Implemented **circuit-breaker + observability layer** — GroundingValidator opens after 3 failures; `GET /metrics` exposes query counts, latency (avg/min/max), and validation skip rate for **production monitoring**.
- Hardened schema validation with length bounds, whitespace sanitisation, semantic emptiness checks, and bounded `top_k` — eliminating a class of silent failure modes before they reach the LLM.

TarangAI | Python, MediaPipe, Stable Diffusion XL, LLaVA-OneVision 7B, OpenCV, Albumentations, React

- Built a **CV + generative AI pipeline** for classical Indian dance: MediaPipe keypoint extraction (33 pose, 42 hand, 468 face pts) per frame + Albumentations augmentation producing **3× dataset expansion** across 6 dance forms.
- Integrated **SDXL** (float16, CUDA) for text-conditioned image generation and **LLaVA-OneVision 7B** for vision-language Q&A over dance imagery; enforced explicit CUDA memory cleanup to prevent OOM across inference sessions.
- Engineered frame extraction pipeline at 256×256px via OpenCV with flip, ±20° rotation, and brightness augmentation — all frames catalogued with dance form, keypoint path, and costume label in `metadata.csv`.

SentimentR | Python, HuggingFace Transformers, LLaMA-2, QLoRA, VADER, NLTK

- Constructed a **dual-path inference engine** in Python — VADER handles low-ambiguity inputs (<5ms), routing to full **LLaMA-2 + CoT** only when confidence falls below threshold, cutting average inference cost without sacrificing precision.
- Curated and annotated a **culturally-tagged sarcasm dataset** with Indian-English idioms and domain-specific labels; applied **QLoRA** for parameter-efficient fine-tuning on consumer GPU memory constraints.
- Developed a **systematic ablation harness** to evaluate routing threshold sensitivity, metadata injection impact, and CoT prompting gain independently — methodology adopted directly into the IEEE paper.

Technical Skills

Languages: Python, SQL, C++, Java **Infra:** FastAPI, Docker, AWS (EC2, S3), Git, CI/CD (GitHub Actions)

AI/ML: LLMs, RAG, Prompt Engineering, Hallucination Mitigation, Confidence Scoring, QLoRA, Stable Diffusion, NLP, Computer Vision, CNNs, RNNs, LSTMs, Transformer Architectures, Ablation Studies

Frameworks: PyTorch, TensorFlow, HuggingFace Transformers, Diffusers, LangChain, FAISS, MediaPipe, OpenCV, spaCy, NLTK, SentenceTransformers, Pandas, NumPy

MLOps: Async Inference, Latency Optimisation, Observability & Metrics, PyTest, Model Versioning, Experiment Tracking

Databases: Qdrant, MySQL

Education

Vellore Institute of Technology Bhopal, Madhya Pradesh

B.Tech – Computer Science Engineering (AIML) – CGPA: 8.5 *Oct 2022 – Aug 2026 (Expected)*

- **Lead, Vitronix Club** — organised technical workshops and led student AI/ML project teams.
- **All India Rank 4**, India Space Lab Internship (Dec 2024 – Jan 2025) — selected nationally for space science and payload analysis.

Certifications

- Prompt Design in Vertex AI – **Google** (May 2024)
- ML & Self-Driving Cars – **Udemy** (Aug 2023)
- AI Job Simulation – **Cognizant** (Sep 2023)